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RESEARCH ARTICLE

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Curcumin-ZnO conjugated nanoparticles confer neuroprotection against ketamine-induced neurotoxicity

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Abstract

Background: Nanotechnology and its application to manipulate herbal compounds to design new neuroprotective agents to manage neurotoxicity has recently increased. Cur-ZnO conjugated nanoparticles were synthesized and used in an experimental model of ketamine–induced neurotoxicity.

Methods: Cur-ZnO conjugated nanoparticles were chemically characterized, and the average crystalline size was determined. Forty-nine adult mice were divided into seven groups of seven animals each. Normal saline was given to control mice (group 1). Ketamine (25 mg/kg) was given to a second group. A third group of mice was given ketamine (25 mg/kg) in combination with curcumin (40 mg/kg), while mice in groups 4, 5, and 6 received ketamine (25 mg/kg) plus Cur-ZnO nanoparticles (10, 20, and 40 mg/kg). Group 7 received only ZnO (5 mg/kg). All doses were ip for 14 days. Hippocampal mitochondrial quadruple complex enzymes, oxidative stress, inflammation, and apoptotic characteristics were assessed.

Results: Cur-ZnO nanoparticles and curcumin decreased lipid peroxidation, GSSG content, IL-1β, TNF-α, and Bax levels while increasing GSH and antioxidant enzymes like GPx, GR, and SOD while increasing Bcl-2 level and mitochondrial quadruple complex enzymes in ketamine treatment groups.

Conclusion: The neuroprotective properties of Cur-ZnO nanoparticles were efficient in preventing ketamine-induced neurotoxicity in the mouse brain. The nanoparticle form of curcumin (Cur-ZnO) required lower doses to produce neuroprotective effects against ketamine-induced toxicity than conventional curcumin.

KEYWORDS

curcumin, ketamine, nanoparticle, neurotoxicity

Abbreviations: ANOVA, analysis of variance; BAX, Bcl-2 associated X-protein; Bcl-2, B-cell lymphoma 2; BSA, bovine serum albumin; CA1, cornu ammonis; CAT, catalase; Cur- ZnO NPs, curcumin- zinc oxide nanoparticles; CYCS, cytochrome c, somatic; **ddH$_2$O**, double distilled water; DG, dentate gyrus; EGTA, ethylene glycol tetraacetic acid; ELISA, enzyme-linked immunosorbent assay; FESEM, field emission scanning electron microscope; FTIR, fourier transform infrared; GPx, glutathione peroxidase; GR, glutathione reductase; GSH, reduced form of glutathione; GSSG, oxidized form of glutathione; **H_2SO_4**, sulfuric acid; IL-1β, interleukin-1 beta; KOH, potassium hydroxide; MDA, malondialdehyde; **$MgCl_2$**, magnesium chloride; **Na_2EDTA**, disodium ethylenediaminetetraacetate dehydrate; **Na_2HPO_4**, sodium phosphate dibasic; NaCl, sodium chloride; **NADH/NAD$^+$**, nicotinamide adenine dinucleotide; NADPH, nicotinamide adenine dinucleotide phosphate; NMDA, N-methyl-D-aspartate; PSA, particle size analyzer; ROS, reactive oxygen species; SDS, sodium dodecyl sulfate; SEM, standard error of the mean; SOD, super oxide dismutase; TBA, thiobarbituric acid; TMB, 3,3′,5,5′ tetramethylbenzidine; TNF-α, tumor necrosis factor alpha; UV-Vis, ultraviolet-visible spectroscopy; XRD, X-ray diffractometer.; ZnO, zinc oxide; **Zn $(NO3)_2$**, zinc nitrate..

1 | INTRODUCTION

Ketamine, an N-methyl-D-aspartate (NMDA) receptor antagonist, induces anesthesia and analgesia.[1] Ketamine, however, can cause hallucinogens and delusions,[1,2] with a high potential for misuse or abuse.[3,4] Clinical and experimental studies have indicated that chronic administration or abuse of ketamine can cause neurobehavioral and neurochemical sequels, promoting neurodegeneration and death of neuronal cells.[1,5] Although the mechanism of ketamine-induced neurodegeneration remains unclear, some studies have suggested that excessive use of ketamine can lead to oxidative stress, inflammation, and apoptosis in neuronal cells, possibly leading to neurobehavioral disorders.[5–7] Ketamine administration can cause anxiety, depression, and cognition impairment in human and animal subjects.[8–10] In addition, mitochondrial dysfunction has been hypothesized as a key mediator of ketamine-induced neurodegeneration.[11–13]

The use of natural substances, including curcumin, in medicine dates back hundreds or even thousands of years.[14] Curcumin, a low molecular weight, lipophilic, yellow polyphenol, is extracted from the rhizomes of the turmeric (*Curcuma longa*) plant.[14] Due to its well-reported biological activity, which includes anti-inflammatory, anti-proliferative, antiangiogenic, proapoptotic, antioxidant, wound healing, anticancer, antiviral, and antidiabetic effects, curcumin has been proposed as an effective therapeutic agent in traditional medicine for various diseases, including cardiovascular, pulmonary, skin, and liver disorders, fatigue, neuropathic pain, bone, muscle loss, and anxiety.[15–17]

Using metal oxide nanoparticles has created numerous medical possibilities by potentially increasing medication delivery.[18] Due to its unique qualities, such as low cost, limited toxicity, abundance in nature, and the capacity to create compounds with various morphologies and structural properties, ZnO, a metal oxide, has a wide range of applications. The use of nanosized ZnO in solar cells[19]; gas sensors, photocatalytic, electrical, and optical devices[20]; electrostatic dissipative coatings; environmental pollutant degradation[21]; and antibacterial agents in lotions, mouthwashes, ointments, and surface coatings to prevent microbial growth are just a few of its numerous applications. The green synthesis of ZnO NPs using *Calotropis procera*, *Aloe barbadensis* miller, and *Poncirus trifoliate* has been reported.[22–25]

The effects of ketamine on the hippocampal region of the adult rodent brain, particularly those associated with chronic use, remain inadequately defined. Based on the prevalence of widespread ketamine addiction, managing the potential neurotoxicity that can result from exposure to this substance necessitates understanding the neurodegenerative effects of ketamine. Our objective was to analyze the potential impact of pretreatment with green synthesized Cur-ZnO NPs on ketamine-induced neurotoxicity in mouse hippocampal tissue and the potential antioxidant, antiapoptotic, and anti-inflammatory effects of ketamine and its impact on mitochondrial enzymes.

2 | MATERIALS AND METHODS

2.1 | Chemicals

Curcumin and KOH were purchased from Merck. Zn $(NO3)_2$ was purchased from Sigma-Aldrich. Ketamine and thiopental sodium were purchased from the Temad Company. The other chemicals were purchased from DNA Biotech Co.

2.2 | Nanoparticle synthesis

Curcumin (0.5 mg) was dissolved in 50 mL of doubly distilled water, and the mixture was heated at 80°C for 48 h. An additional 50 mL of doubly distilled water was added, followed by adding 0.1 M Zn $(NO3)_2$. The resulting yellowish solution was refluxed for 2 h at 85°C–90°C for 2 h. After cooling, 5 mL of 0.2 M KOH was slowly added to the solution in an ice bath. The gel-like orange-yellow suspension was centrifuged at 1957*g* for 10 min. The supernatant fraction was decanted until there was no yellow tint, and the precipitant was washed with doubly distilled water, followed by an acetone wash before a final water rinse to remove the remaining unreacted curcumin. The Cur-ZnO NPs precipitate was vacuum-dried at room temperature and stored in the dark at 4°C.

2.3 | Pharmacological study

2.3.1 | Animals

Forty-nine BALB/c male mice (25–30 g) were purchased from the Experimental Research Center of Iran University of Medical Sciences. Animals were housed in groups of seven in wire cages in the animal facility of the Experimental Research Center at Masih Danshvari Hospital in Shahid Beheshti University of Medical Sciences. Mice had free access to pelleted feed (Pars Animal Feed Co) and urban drinking water. The room temperature was 22 ± 0.5°C, and the relative humidity was 6–40-% with a 12-h light/dark cycle.

2.3.2 | Cur-ZnO NPs effects against ketamine-induced neurotoxicity

Ketamine, curcumin, ZnO, and Cur-ZnO nanoparticles were dissolved in normal saline. The dose selection for ketamine, curcumin, and Cur-ZnO nanoparticles was based on the literature.[1,5,26–31] Treatment was daily for 14 consequent days by intraperitoneal injection. Mice in the control group[1] were treated with 0.2 mL of normal saline. Group two[2] animals received ketamine (25 mg/kg). Group three animals[3] were treated with curcumin (40 mg/kg) and ketamine (25 mg/kg). In groups four, five, and six, mice received Cur-ZnO nanoparticles at 10, 20, and 40 mg/kg, respectively. Group seven animals received ketamine (25 mg/kg) and ZnO (5 mg/kg). A schematic of the protocol and the experimental timeline is shown in Figure 1.

On Day 15, animals were anesthetized with thiopental (50/mg/kg, i.p.), euthanized, and necropsied. Brains were carefully removed, and the hippocampus was isolated and perfused.[32,33] The hippocampus's right hemisphere was evaluated for oxidative stress, inflammation, apoptosis, and mitochondrial respiratory chain enzyme activity. These hippocampal samples were stored at −70°C.[34] Histopathology was conducted on the left hemisphere of the hippocampus.

2.3.3 | Protein concentration

Hippocampus tissue was homogenized in a cold homogenization buffer (25 mM 4-morpholinepropanesulfonic acid, 400 mM sucrose, 4 mM $MgCl_2$, and 0.05 mM EGTA, pH, 7.3) and centrifuged at 450*g* for 10 min. The supernatant fraction was isolated, centrifuged for 10 min at 12,000*g*, mixed with the homogenization buffer, and stored at 0°C until analyzed. The hippocampal tissue homogenate (5, 10, 15, 20, 25, and 30 µL) and 100 µL of the Bradford reagent (DNA Biotech Co) were added to each well of a multiwell plate. The plate was gently shaken, and the optical density was measured at 630 nm. Bovine serum albumin (DNA Biotech Co.) was dissolved in the homogenization buffer and serially diluted (0.1–1.0 mg/mL) for the standard curve. Total protein was estimated by the Bradford method.[35,36]

2.3.4 | Oxidative stress parameters

Lipid peroxidation

The primary byproduct of cellular lipid peroxidation is malondialdehyde (MDA). One hundred microliters of the MDA standard (DNA Biotech Co. 100 uL) or 100 uL of the tissue homogenate was added to wells of the 96 microwell plate. SDS Lysis Solution (100 UL) was added to each well. TBA reagent (250 uL) (DNA Biotech Co.) was added, and the mixture was gently shaken and incubated at 95°C for 45–60 min. The samples were then centrifuged at 1000*g* for 15 min. n-Butanol (DNA Biotech Co.) (300 uL) was added to stop the reaction. Samples were centrifuged at 10,000*g* for 7 min, and the absorbance was read at 532 nm. Results are reported as nmol/mg of protein.[37–42]

Glutathione (GSH) and oxidized GSH (GSSG)

Twenty-five microliters of the glutathione reductase solution (1X) (DNA Biotech Co.) and 25 uL of the NADPH solution (1X) (DNA Biotech Co.) were added, respectively, to 96-well plates to measure GSH and GSSG content. A standard glutathione solution (DNA

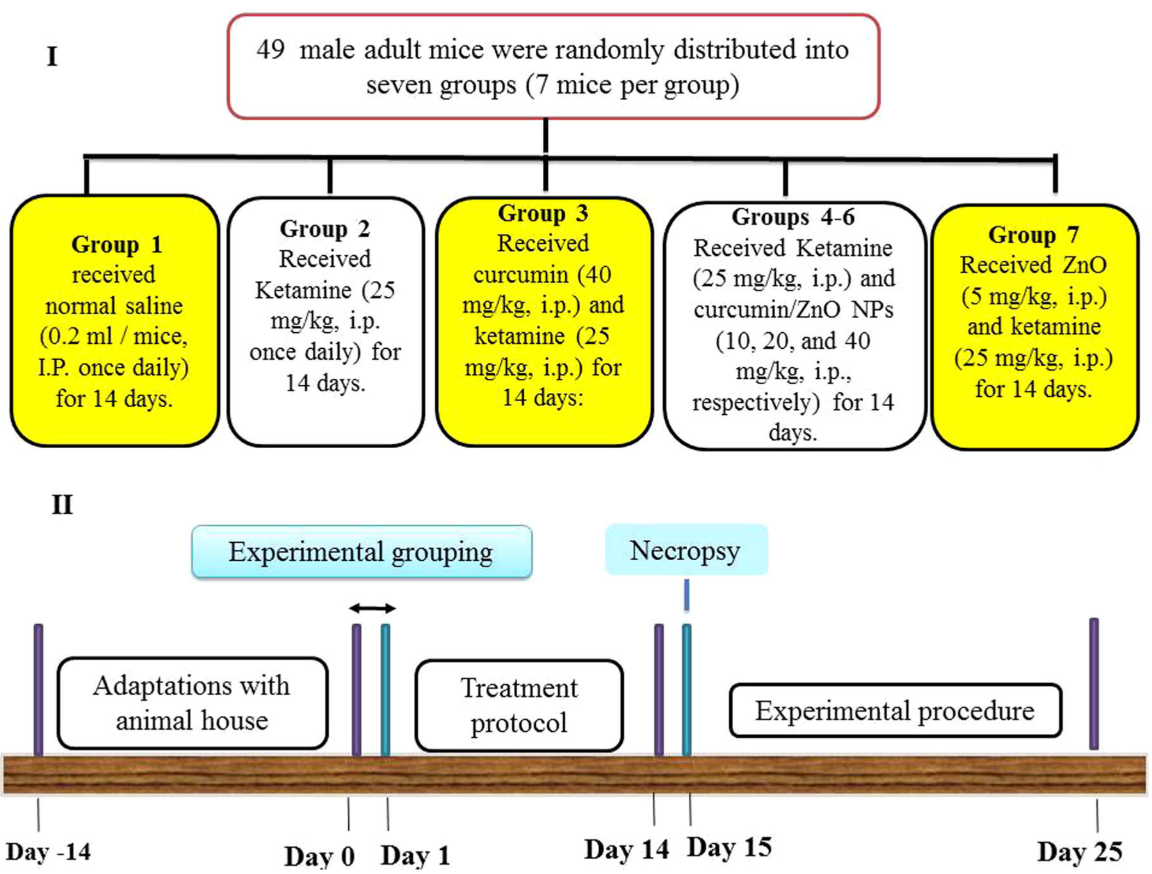

FIGURE 1 Schematic Illustration of experimental grouping (I) and timeline for the experimental procedure (II).

Biotech Co.) or 25 uL homogenized tissue sample was added. Chromogen (1X) (50 uL, DNA Biotech Co.) was added, and the mixture was vigorously blended. Absorbance was measured at 405 nm. The results are reported as nmol/mg of protein.[42,43]

Super oxide dismutase (SOD)

SOD activity was measured using a kit supplied by DNA Biotech Company (DNA Biotech Co). Twenty microliters of ddH_2O was added to the first and third blank wells, and 20 μL of homogenized tissue solution was added to the second blank wells. Then, 200 μL of the working solution (water-soluble tetrazolium salt) was added to each well and mixed gently. Twenty microliters of the dilution buffer was added to the second and third wells, and 20 μL of the enzyme working solution was added to the first well and the sample blank. The solutions were mixed vigorously and incubated at 37°C for 20 min. Absorbance was recorded at 450 nm. SOD activity is reported as units/mL/mg protein.[42–44]

Glutathione peroxidase (GPx)

A commercial DNA Biotech Company kit (DNA Biotech Co.) was used to measure GPx activity in the hippocampal tissue. Twenty microliters of the homogenized tissue solution plus the assay buffer and 20 μL of the assay buffer were added to separate wells. Two hundred microliters of the reaction solution and 20 μL of the peroxidase substrate solution were mixed into each well, and absorbance was read at 340 nm over 8 min at 25°C. Results are reported as mU/mg protein.[42–44]

Glutathione reductase (GR)

GR activity was measured by following the conversion of GSSG to GSH in the presence of NADPH using a DNA Biotech Company kit (DNA Biotech Co.). The reaction mixture included 200 μM of potassium phosphate buffer (pH = 5.7), 50 μM of NADPH, 500 μM of GSSG, 1.5 μM of $MgCl_2$, 0.2 μM of Na_2EDTA and 50 μL of tissue homogenized sample. The mixture was vigorously mixed, and absorbance was read at 340 nm for 120 s. Results are expressed as mU/mg protein.[42–44]

2.3.5 | Mitochondrial complex enzymes

Mitochondria complexes I, II, III, and IV activities were measured using commercial kits (Abcam, Co.). Mitochondrial complex I activity was estimated based on the oxidation of NADH to NAD^+, and its absorbance was read at 450 nm. Mitochondrial complex II was assessed according to catalysis of the electron transfer of succinate to ubiquinone, and its absorbance was read at 550 nm. The reaction speed for the conversion of the oxidized form of CYCS to its reduced form at 600 nm was used to estimate mitochondrial complex III activity. Measurement of the oxidation of the reduced form of CYCS at 550 nm was used to monitor mitochondrial complex IV activity. Reaction velocities (V max) were calculated from the most linear part of the output curve. All values are reported as activity/mg of protein/min.[34,45]

2.3.6 | TNF-α, IL-1β, Bax, Bcl-2, caspase-3, and caspase-7

TNF-α, IL-1β, Bax, Bcl-2, caspase-3, and caspase-7 were estimated in hippocampal tissue using commercial ELISA kits (Genzyme Diagnostics). The sensitivity and specificity of the TNF-α, IL-1β, Bax, Bcl-2, caspase-3, and caspase-7 kits were <5 ng/mL and 76%, respectively. Each sample's optical density (OD) was obtained with an ELISA reader using the ELISA protocol, and the OD was inserted into the generated standard curve to obtain the protein concentration. Sheep anti-mouse IL-1β, TNF-α Bax, Bcl-2, caspase-3, and caspase-7 polyclonal antibodies (Sigma Chemical Co.) were washed three times (0.5 M sodium chloride (NaCl), 2.5 mM sodium dihydrogen phosphate (NaH_2PO_4), 7.5 mM Na2HPO4, 0.1% Tween 20, pH 7.2). One hundred milliliters of 1% (w/v) ovalbumin solution (Sigma Chemical Co.) was added to each well and incubated at 37°C for 1 h. After washing three times, 100 mL of sample or standard was added to each well and incubated at 48°C for 20 h. After three washes, 100 mL of biotinylated sheep anti-mouse IL-1β or TNF-α antibody (1:1000 dilutions in wash buffer containing 1% sheep serum, Sigma Chemical Co.) was added to each well. Following

TABLE 1 Effects of Cur-ZnO NPs on hippocampal GSH and GSSG content in ketamine-treated mice.

Group	Mean ± SEM GSH (nmol/mg protein)	GSSG (nmol/ mg protein)	GSH/ GSSG
Control group	121.6 ± 7.2	0.96 ± 0.5	120
Ketamine (25 mg/kg)	18.2 ± 4.1[a]	22.9 ± 2.1[a]	0.8[a]
Ketamine + curcumin (40 mg/kg)	112.4 ± 6.9[b]	3.3 ± 0.9[b]	33[b]
Ketamine + Cur-ZnO nanoparticles (10 mg/kg)	82.5 ± 7.6[b]	6.2 ± 0.6[b]	13[b]
Ketamine + Cur-ZnO nanoparticles (20 mg/kg)	111.3 ± 9.6[b]	3.6 ± 0.3[b]	30[b]
Ketamine + Cur-ZnO nanoparticles (40 mg/kg)	114.5 ± 8.3[b]	2.3 ± 0.4[b]	49.7[b]
ZnO (5 mg/kg)	43.5 ± 16.9	16.6 ± 4.3[a]	2.6[b]

Note: All data are mean ± SEM, $n = 7$.

[a]Significant level with $p < 0.001$ vs. the control group.

[b]Significant level with $p < 0.001$ ketamine (25 mg/kg).

1 h of incubation and three washes, 100 mL of Avidin-HRP (Dako Ltd) (1:5000 dilutions in wash buffer) was added to each well, and the plates were incubated for 15 min. After washing three times, 100 mL of TMB substrate solution (Dako Ltd.) was added to each well and incubated for 10 min at room temperature. Finally, 100 mL of 1 M H_2SO_4 was added to stop the reaction, and the absorbance was read at 450 nm. Results are expressed as ng IL-1β/mL or TNF-α/mL or Bax and Bcl-2, caspase-3, and caspase-7.[46,47]

2.3.7 | Pathology

The vascular system was perfused with 4% paraformaldehyde at necropsy.[32,46] The hippocampus tissue was washed with physiological saline, fixed in paraformaldehyde, 10% w/v, then dehydrated using a series of ethanol steps and embedded in paraffin. Five micrometres thick sections were mounted on slides and stained with hematoxylin and eosin. Images of 20 segments per group (×400 and 200 magnifications) were captured and evaluated using morphometry software (Optikavision Pro). The cell density and changes in cell shapes were determined in a region of 1.30 mm of the hippocampal subfield.[47–49]

2.3.8 | Statistical analysis

Graph Pad PRISM v.6 Software collected and analyzed all data. Mean ± standard error (SEM) was calculated. ANOVA and Tukey's posttest assessed the significant differences between the control and

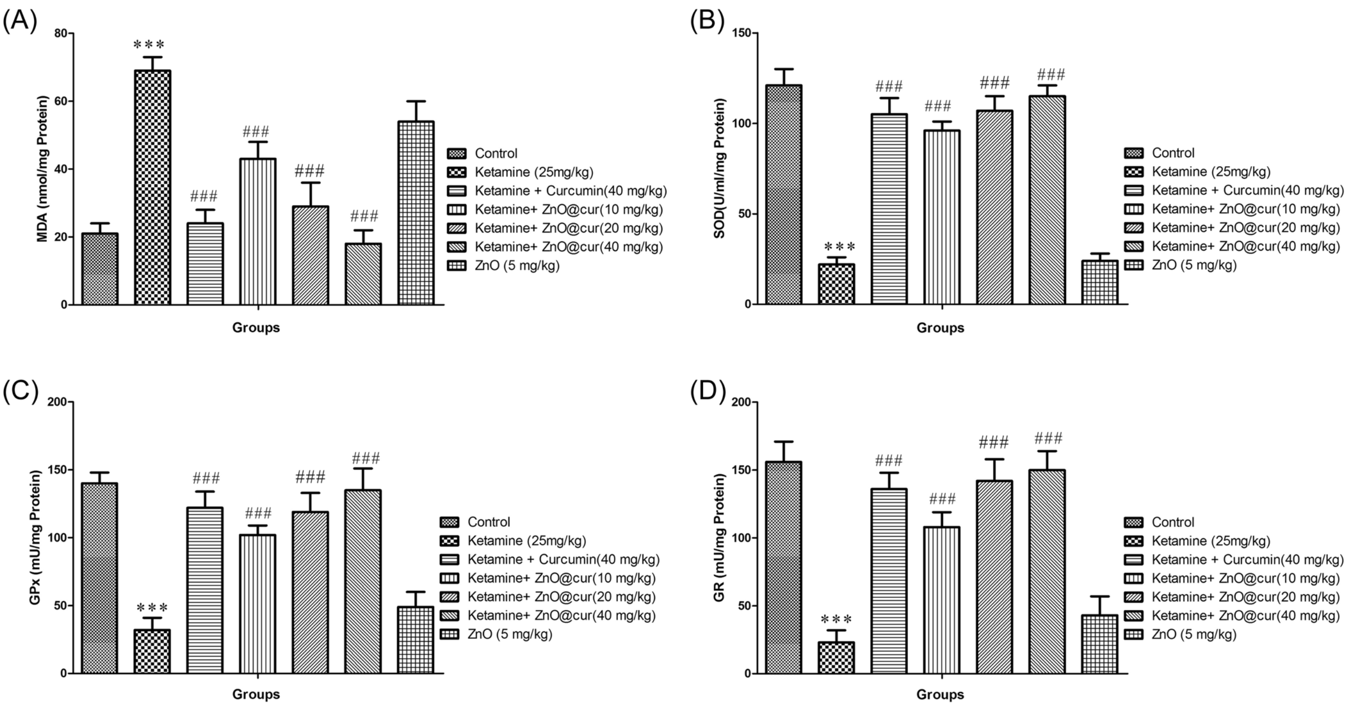

FIGURE 2 Effects of Cur-ZnO nanoparticles on ketamine-induced alteration in lipid peroxidation (A), SOD activity (B), GPx activity (C), and GR activity (D) in mouse isolated hippocampus. All data are mean ± SEM (n = 7). ***p < 0.001 versus control ###p < 0.001 versus ketamine (25 mg/kg).

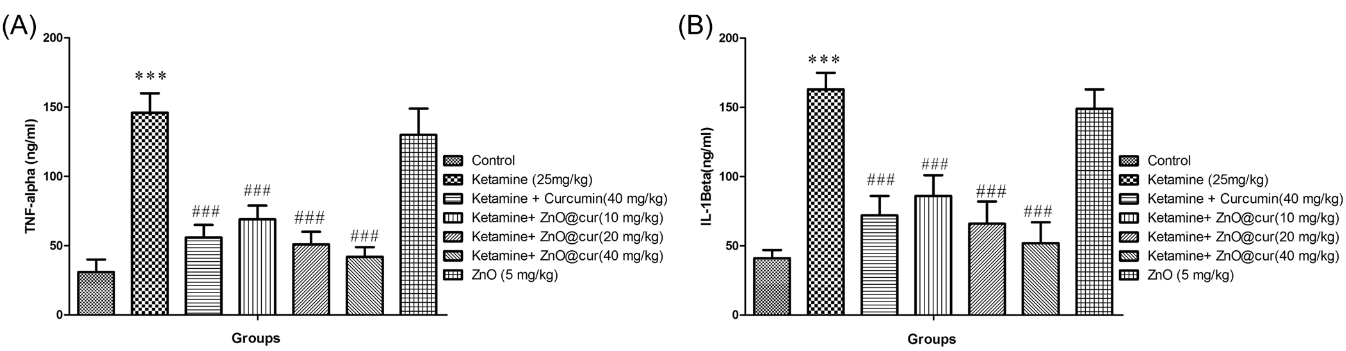

FIGURE 3 Effects of Cur-Zno nanoparticles on ketamine induced alteration in TNF-A (A) and IL-1β (B) levels in mouse isolated hippocampus. All data are mean ± SEM (n = 7). ***p < 0.001 versus control ###p < 0.001 versus ketamine (25 mg/kg).

treated groups. The Pearson's correlation analysis was used for the correlation of each TNF-α or IL-1β expression with the mitochondrial quadruple complex activities or TNF-α or IL-1β expression with Bax, Bcl-2, Beclin-1, and caspase-3. $p < 0.05$ was considered significant for correlation. Also, for all parameters in all experiments in this study an F test with[6,42] degrees of freedom was reported. In the results section the number in parentheses after each experimental parameter is the F ratio followed by the p value.

3 | RESULTS

3.1 | Cur-ZnO nanoparticles

The chemical and physical characteristics of the Cur-ZnO nanoparticles were determined using XRD and UV-visible analysis, FTIR analysis, and FESEM analysis (Supporting Information: 1).

3.2 | Effect of Cur-ZnO NPs on ketamine-induced GSH/GSSG

Ketamine decreased GSH (18.89; $p < 0.001$) while increasing GSSG (20.07; $p < 0.001$) compared to the control (Table 1). The effects of ketamine treatment improved in mice treated with ketamine and curcumin and with ketamine + nanoparticles of Cur-Zn. In mice receiving ketamine plus 40 mg/kg curcumin or 40 mg/kg NP Cur-Zn, the recovery in both GSH (18.89; $p < 0.001$) and GSSG (20.07; $p < 0.001$) approached control values. Treatment of animals with ZnO alone decreased GSH and increased GSSG, but neither was statistically significant.

3.3 | Effects of Cur-ZnO NPs on ketamine-induced oxidative stress

Ketamine decreased SOD (39.82; $p < 0.001$), GPx (13.96; $p < 0.001$), and GR (16.71; $p < 0.001$) but increased MDA (15.36; $p < 0.001$) compared to control mice (Figure 2A–D). Ketamine and curcumin (40 mg/kg) or nanoparticles of Cur-ZnO reversed the effects of ketamine alone (Figure 2A–D). The results of ZnO (5 mg/kg) treatment paralleled ketamine (25 mg/kg) treatment (Figure 2A–D).

3.4 | Effect of Cur-ZnO NPs on ketamine-induced inflammation

Ketamine (25 mg/kg) significantly increased IL-1β (12.39; $p < 0.001$) and TNF-α (14.83; $p < 0.001$) compared to controls (Figure 3A,B). IL-1β (12.39; $p < 0.001$) and TNF-α (14.83; $p < 0.001$) levels were decreased in the ketamine combined with curcumin (40 mg/kg) group compared to animals treated with ketamine alone (Figure 3A,B). The nanoparticles of Cur-ZnO inhibited the effect of ketamine-induced inflammation and the levels of IL-1β (12.39; $p < 0.001$) and TNF-α (14.83; $p < 0.001$) at all doses compared to

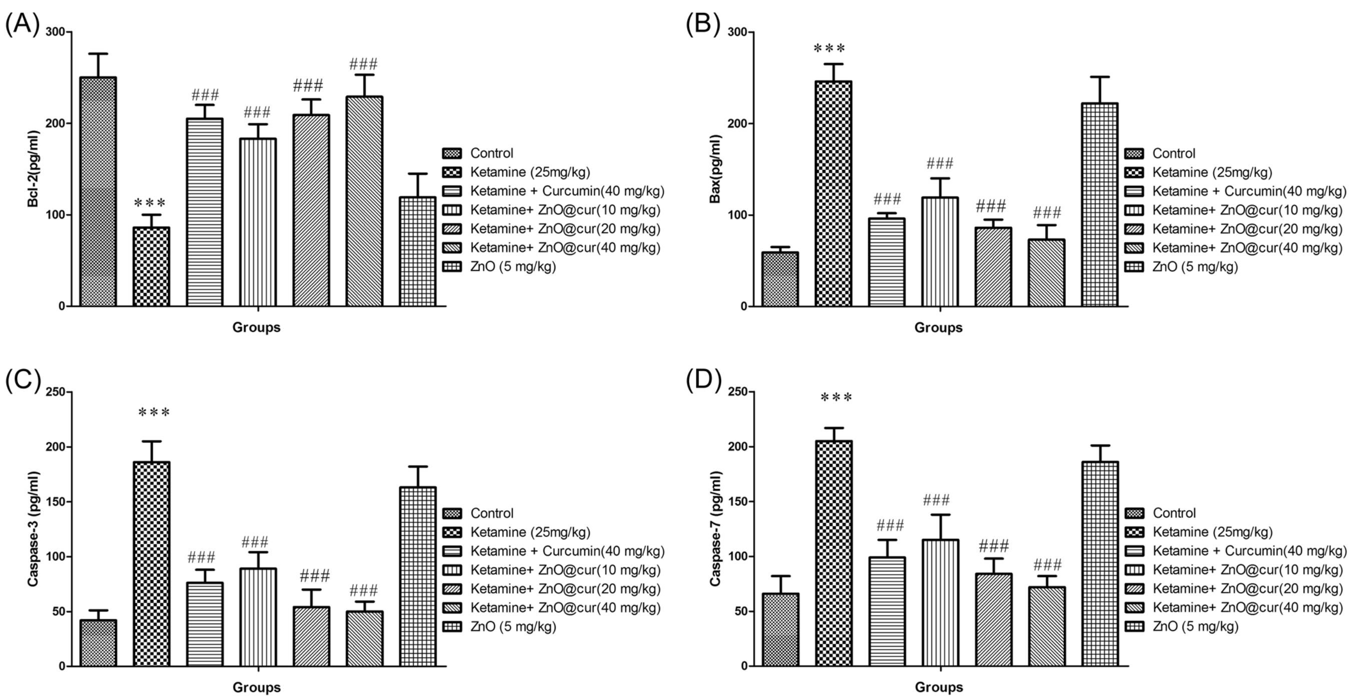

FIGURE 4 Effects of Cur-Zno nanoparticles on ketamine induced alteration in protein expression of Bcl-2 (A), Bax (B), caspase-3 (C), and caspase-7 (D) in mouse isolated hippocampus. All data are mean ± SEM ($n = 7$). $^{***}p < 0.001$ versus control $^{\#\#\#}p < 0.001$ versus ketamine (25 mg/kg).

animals treated with ketamine alone (Figure 3A,B). ZnO (5 mg/kg) paralleled ketamine (25 mg/kg) treatment (Figure 3A,B).

3.5 | Effects of Cur-ZnO NPs on ketamine-induced apoptosis

Ketamine (25 mg/kg) significantly increased Bax (19.00; $p < 0.001$), caspase-3 (15.32; $p < 0.001$), and caspase-7 (12.68; $p < 0.001$) levels and decreased Bcl-2 (8.575; $p < 0.001$) compared to controls (Figure 4A–D). The combination of ketamine and curcumin increased Bcl-2 (8.575; $p < 0.001$) but decreased Bax (19.00; $p < 0.001$), caspase-3 (15.32; $p < 0.001$), and caspase-7 (12.68; $p < 0.001$) compared to ketamine-only treated mice (Figure 4A–D). Nanoparticles of Cur-ZnO in all doses suppressed the effect of ketamine-induced apoptosis by enhancing Bcl-2 (8.575; $p < 0.001$) and decreasing Bax (19.00; $p < 0.001$), caspase-3(15.32; $p < 0.001$), and caspase-7 (12.68; $p < 0.001$) levels compared to ketamine-only treated mice (Figure 4A–D). Additionally, although administering ZnO to mice prevented some of the apoptosis caused by ketamine (25 mg/kg), these effects were not statistically significant (Figure 4A–D).

3.6 | Effect of Cur-Zn0 NPs on ketamine-induced mitochondrial chain enzyme

Ketamine (25 mg/kg) significantly reduced the activity of mitochondrial complexes I (5.991; $p < 0.001$), II (12.11; $p < 0.001$), III (3.418; $p < 0.001$), and IV (3.781; $p < 0.001$) compared to the controls (Figure 5A–D). Ketamine combined with curcumin (40 mg/kg) enhanced the activity of all four mitochondrial complexes compared to the ketamine-treated animals (Figure 5A–D). The activity of the mitochondrial complex I (5.991; $p < 0.001$), II (12.11; $p < 0.001$), III (3.418; $p < 0.001$), and IV (3.781; $p < 0.001$) enzymes was increased in comparison to the ketamine-only treated animals at all doses exposed to the nanoparticles of Cur-ZnO (Figure 5A–D). ZnO did not alter the changes induced by ketamine (Figure 5A–D).

3.7 | Histopathological changes

Ketamine (25 mg/kg) significantly induced cell degeneration and shrinkage and decreased cell density of granular cells in the DG (11.38; $p < 0.001$) area and the pyramidal cells in the CA1 (9.811; $p < 0.001$) area compared with controls (Figures 6 and 7). Both

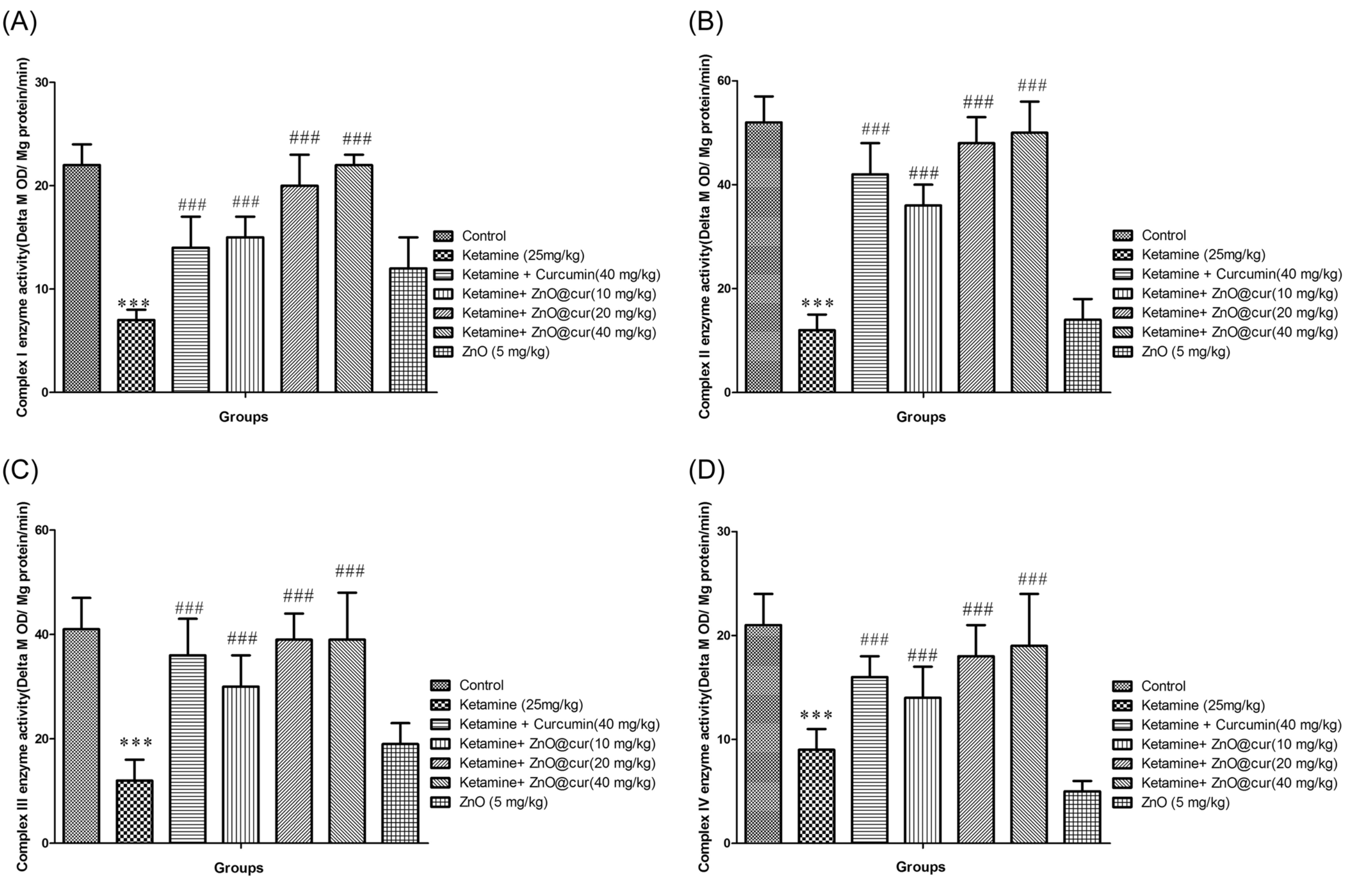

FIGURE 5 Effects of Cur-Zno nanoparticles on ketamine induced alterations on mitochondrial complex enzymes activity. complex I (A), complex II (B), complex III (C), and complex III (D) in mouse isolated hippocampus. All data are mean ± SEM ($n = 7$). $^{***}p < 0.001$ versus control $^{\#\#\#}p < 0.001$ versus ketamine (25 mg/kg).

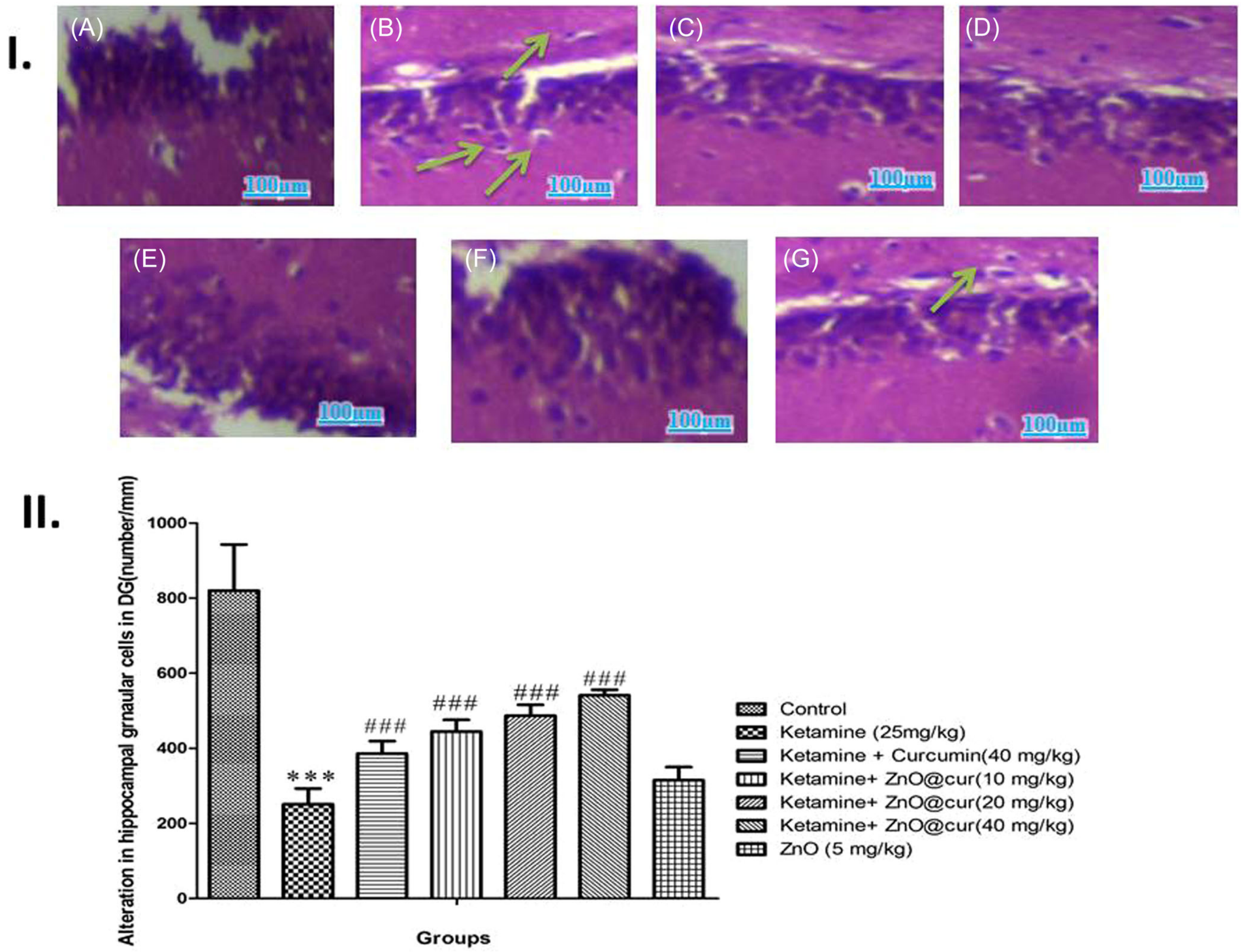

FIGURE 6 (I) Representative images of hematoxylin and eosin-stained in granular cells Of DG. control group (A), under treatment with only ketamine (25 Mg/Kg) (B), under treatment with ketamine (25 Mg/Kg) and Curcumin (40 Mg/Kg) (C), under treatment with ketamine (25 Mg/Kg) and Cur-ZnO NPs with doses of 10, 20, and 40 mg/kg, respectively (D–F), under treatment with ketamine (25 mg/kg) and ZnO (5 mg/kg) (G). Green arrow indicates the vacuolation and neuronal degeneration in the DG area of the mouse hippocampus (magnification ×200. Scale bar 100 μm). (II) Effects of Cur-ZnO nanoparticles on ketamine induced in cell density (counts) of granular cells Of DG. All data are mean ± SEM ($n = 7$). $^{***}p < 0.001$ versus control $^{\#\#\#}p < 0.001$ versus ketamine (25 mg/kg).

combinations (ketamine + curcumin and ketamine + Cur-ZnO nanoparticle) reversed the histopathology observed in ketamine-only treated mice (Figures 6 and 7). ZnO did not display the histomorphological changes observed in ketamine-treated mice (Figures 6 and 7).

4 | DISCUSSION

Ketamine is an analgesia and anesthesia used in obstetric and pediatric patients.[13,50] Ketamine, however, can cause hallucinogens and delusions[51] and has a significant potential for abuse.[4] Ketamine dramatically reduced the survival rate of neuron cell lines in a time and dose-dependent manner.[52,53] Although its mechanism remains unclear, ketamine can induce neurodegeneration and neuronal cell death,[11–13] specifically damaging granular and glial cells in the hippocampus.[11–13] Both oxidative stress and a decline in antioxidant levels have been observed following ketamine treatment.[54,55] During oxidative stress, reactive oxygen species (ROS), including hydroxyl radicals, are produced, stimulating DNA oxidation, resulting in 8-OHG, replication errors, and mutations.[52,56] Furthermore, ketamine-induced inflammation and a mutual feed-forward connection between oxidative stress and inflammation may result in DNA damage and apoptosis.[56] Both curcumin and Zn0 curcumin NPs were effective in reversing the untoward effects induced by ketamine in the mouse hippocampus; therefore, both may be potentially effective therapeutic agents (Figure 8).

Curcumin (diferuloylmethane), the most abundant bioactive component extracted from the rhizomes of *Curcuma longa*,[57–59] is a dietary phenol with medical and pharmacological properties.[60–62] Curcumin has antioxidant, anti-inflammatory, antiapoptotic, and immunomodulatory properties[63] and has been used to treat

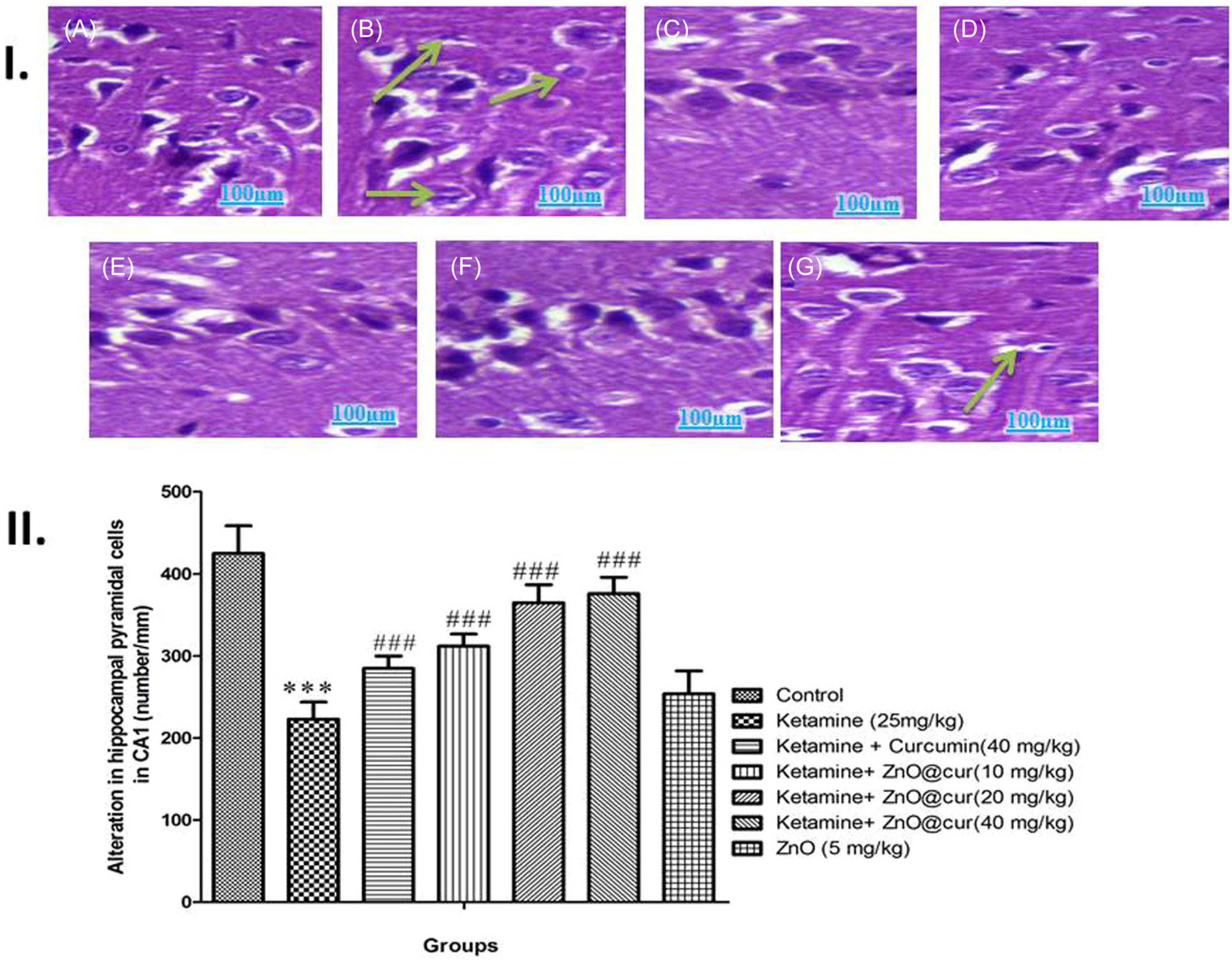

FIGURE 7 (I) Representative images of hematoxylin and eosin-stained pyramidal cells of the CA1 area. Control group (A), under treatment with only ketamine (25 mg/kg) (B), under treatment with ketamine (25 mg/kg) and curcumin (40 mg/kg) (C), under treatment with ketamine (25 mg/kg) and Cur-ZnO NPs with doses of 10, 20, and 40 mg/kg, Respectively (D–F), under treatment with ketamine (25 mg/kg) and ZnO (5 mg/kg) (G). Green arrow indicates the vacuolation and neuronal degeneration in the CA1 area of the rat hippocampus (magnification ×200. Scale Bar 100 μm). (II) Effects of Cur-ZnO nanoparticles on ketamine induced cell density (counts) in the pyramidal cells of the CA1 area (magnification ×200). All data are mean ± SEM ($n = 7$). $^{***}p < 0.001$ versus control $^{###}p < 0.001$ versus ketamine (25 mg/kg).

neurodegenerative diseases such as Alzheimer's and Parkinson's.[39,60,64,65] Curcumin reduced lipid peroxidation, improved the activity of antioxidant enzymes such as SOD and CAT, and increased glutathione-related antioxidants.[66,67] Similarly, curcumin lowered the alcohol-induced increase of TNF-α, IL-1β, and TGF-1 levels.[68–70] Curcumin promotes neuroprotection, at least in part, by scavenging free radicals.[71] Curcumin has been proposed for treating neurodegenerative diseases due to its ability to boost glutathione synthesis.[14]

Numerous studies have demonstrated that nanotechnology can improve the therapeutic effectiveness of curcumin as well as its pharmacokinetics and pharmacodynamic properties by altering its structure and thereby increasing its absorption [90, 93] and providing the potential of delivering it to specific cells and tissues such as the brain.[72,73] An additional goal of targeted curcumin delivery as an NP is to prevent or at least reduce side effects while lowering consumption and treatment costs.[72,74]

The current work demonstrated that treatment with curcumin-conjugated ZnO nanoparticles (Cur-ZnO nanoparticles) and curcumin prevented or reversed ketamine-induced neurodegeneration in the mouse hippocampus. Cur-ZnO nanoparticles protected against ketamine-induced mitochondrial enzyme failure and its repercussions in the adult mouse hippocampus. The hippocampus in mice treated with ketamine demonstrated more significant levels of apoptotic biomarkers such as Bax, caspase-3, and caspase-7 and higher levels of inflammatory cytokines such as TNF-a and IL-1 than control mice. Ketamine lowered Bcl-2. Ketamine reduced the activity of the quadruple enzymes of the mitochondrial respiratory chain. Cur-ZnO nanoparticles decreased oxidative stress and neuroinflammation in ketamine-treated mice in a dose-dependent manner. Cur-ZnO

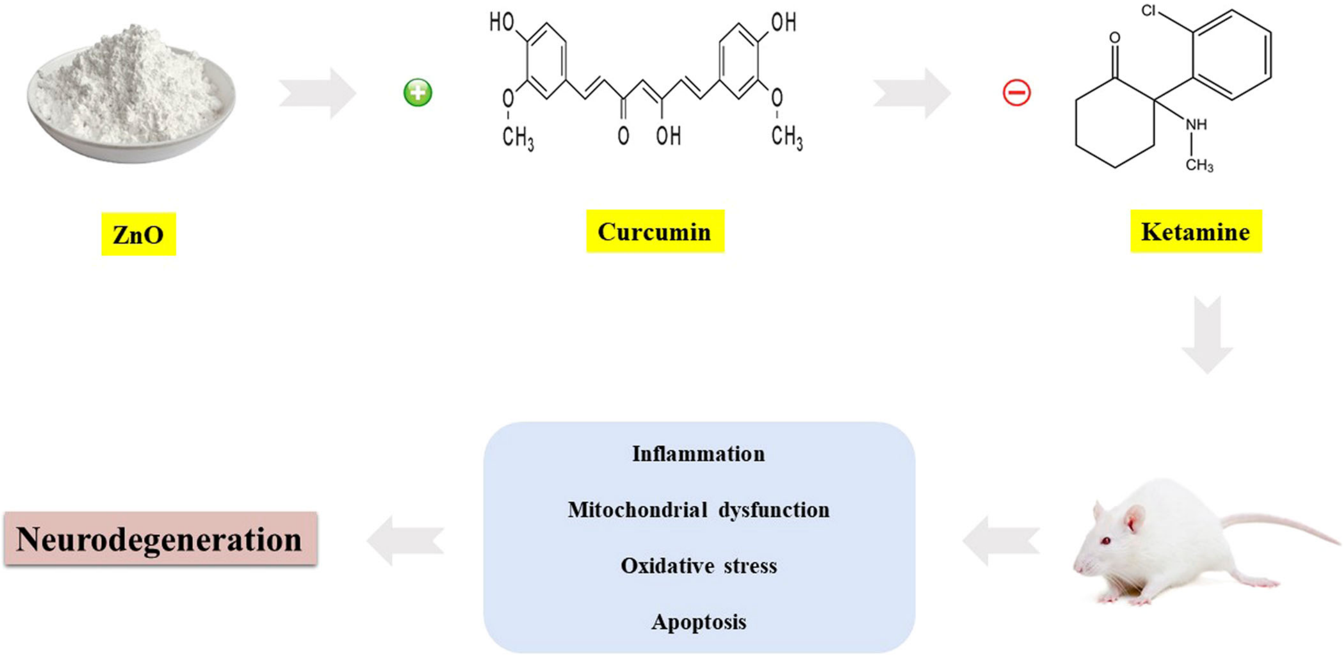

FIGURE 8 Curcumin/ZnO NPs are an effective neuroprotective agent against oxidative stress, inflammation, and apoptosis caused by ketamine.

nanoparticles also decreased apoptosis-induced neuronal cell death and enhanced the activity of quadruple mitochondrial respiratory chain enzymes in the ketamine-treated animals.

Curcumin alone blocked ketamine-induced neuro-inflammation and apoptosis and reduced IL-1β, TNF-α, and Bax while increasing Bcl-2, which is consistent with previous studies demonstrating that curcumin can block multiple sites of TNF-α, TGF-β, and apoptosis signaling cascades, thereby protecting the brain from inflammation and apoptosis damage caused by drug misuse.[17,37,39]

Curcumin alone enhanced GSH content and decreased GSSG levels in mice treated with ketamine, in addition to reducing MDA levels. Cur-ZnO nanoparticles significantly accelerated the action of key antioxidant enzymes such as SOD, GR, and GPx in ketamine-treated rats. Furthermore, curcumin reduced ketamine's adverse effects on these antioxidant enzymes. As a result of improving the GSSG/GSH ratio, even modest dosages of Cur-ZnO nanoparticles and intermediate doses of curcumin improved the transformation of GSSG to GSH and protected the brain from ketamine-induced oxidative stress. Previous research[17,37,39] has shown that curcumin has a high potential for modifying the oxidant-antioxidant system in drug users.

Ketamine caused cell death, reduced cell density, and enhanced granular cell degenerative changes in both the granular and pyramidal cell layer in the D.G. and CA1 area of the hippocampus.[75–77] These histomorphological changes were probably mediated via activation of oxidative stress, mitochondrial dysfunction, inflammation, and apoptosis pathways.[50,78,79] Curcumin conjugated ZnO nanoparticles, and curcumin reduced these ketamine–induced histomorphological changes.

5 | CONCLUSION

A green synthesis approach successfully synthesized the curcumin-linked ZnO nanostructure used in these studies (Supporting Information: 1). Both curcumin and Cur- ZnO nanoparticles reversed the untoward effects of ketamine in the mouse hippocampus. Curcumin and the Cur- ZnO nanoparticles protected against ketamine-induced oxidative stress, mitochondrial dysfunction, inflammation, apoptosis, and hippocampal histopathological changes. The nanoparticle form of curcumin (Cur-ZnO) required lower doses to produce neuroprotective effects against ketamine-induced toxicity than conventional curcumin. Both, but especially the Cur- ZnO nanoparticles, may potentially reduce the neurodegenerative and neurotoxic side effects of ketamine, especially when ketamine is used chronically or abused.

AUTHOR CONTRIBUTIONS

Conception and design: Majid Motaghinejad, Malak Hekmati, Fatemeh Mobinhosseini, and Mahsa Salehirad. *Acquisition of data*: Majid Motaghinejad, Malak Hekmati, Fatemeh Mobinhosseini, and Mahsa Salehirad. *Interpretation of data*: Majid Motaghinejad, Malak Hekmati, Fatemeh Mobinhosseini, and Mahsa Salehirad. *Drafting the manuscript or revising it critically for important intellectual content*: Majid Motaghinejad, Sepideh Safari, and Mina Gholami. *Final approval of the version to be published*: Majid Motaghinejad and A. Wallace Hayes. *Participated sufficiently in the work to take public responsibility for appropriate portions of the content*: Majid Motaghinejad and A. Wallace Hayes. *Agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of* *any part of the work are appropriately investigated and resolved*: Fatemeh Mobinhosseini, Mahsa Salehirad, Wallace Hayes, Majid Motaghinejad, Malak Hekmati, Malak Hekmati, and Mina Gholami.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

All data (Mean±SEM) is available.

ETHICS STATEMENT

The protocol followed the Guidelines of Animal Ethics and Welfare, based on the ARRIVE procedures.[80] The protocol was approved by the animal use and care committee of the Shahid Beheshti University of Medical Sciences [Protocol and Ethical Code Number: I.R.SBMU.NRITLD.REC.1402.93].

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Additional supporting information can be found online in the Supporting Information section at the end of this article.

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